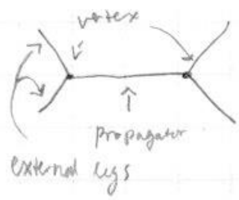
Quantum Field Theory 2 Homework 7

Exercise 7.8

$$\mathcal{L} = \underbrace{\frac{1}{2}(\partial_\mu \phi_R)^2 - \frac{1}{2}m^2\phi_R^2 - g\frac{\mu^{\varepsilon/2}}{3!}\phi_R^3}_{=\mathcal{L}_R} + \underbrace{\frac{\delta_Z}{2}(\partial_\mu\phi_R)^2 - \frac{\delta_m}{2}\phi_R^2 - \frac{\delta_g \mu^{\varepsilon/2}}{3!}\phi_R^3 + \delta_1\phi_R}_{=\mathcal{L}_{CT}}$$

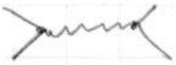
We are working in six dimensions: $d = 6-\varepsilon$

let's write: p = # of internal vertices, n = # of external lines, v = # of vertices.



Each external leg connects to on vertex and propagator to two vertex and a vertex to 2-external legs and a single propagator:

$\Rightarrow$ $3v = 2p + n$ must be satisfied

for example  has $v=2$, $n=4$, $p=1$

$\Rightarrow$ $6 = 2\cdot 1 + 4 = 6$ □ and

The number of unconstrained loop integral is

$L = p - v + 1$ // each internal momentum integrated over, vertex includes one constraint on momentum conservation, overall momentum conservation gives +1

$\Rightarrow$ Each integral adds 6 powers of momentum to the numerator $(d = 6-\varepsilon)$ (six dimensional) and every propagator adds 2 powers of momentum to denominator.

$\Rightarrow$ The degree of divergence is $D = 6L - 2p = 6 + \underbrace{4p - 6v}_{=-2n} = 6 - 2n$

$\Rightarrow$ $D = 6-2n$ means D is divergent for $n \leq 3$

$\Rightarrow$ 3-point function has $n=3$ $\Rightarrow$ $D=0$ $\Rightarrow$ divergent

$\Rightarrow$ 2-point function and 1-point function are also divergent since then we have $D \geq 0$

$\Rightarrow$ 1-, 2- and 3-point function are divergent.

We start with 2-point function:

The two point function is  $= G(p^2) = \dfrac{i}{p^2 - m^2 - \Pi(p) + i\varepsilon}$, using the geometric progression of the loops.

with $\quad -i\Pi(p^2) = \underbrace{-i\,\Pi^{(1)}(p^2)}_{\text{1-loop correction}} + \underbrace{i(\delta_Z\, p^2 - \delta_m)}$

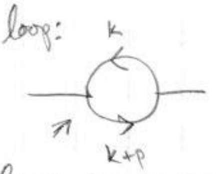
The term from counter term Lagrangian via the feynman rule.

1-loop
Counter term:

 : $\; i(\delta_Z\, p^2 - \delta_m)$ is the two point function? of counter term (Peskin page 325)

The 1-loop get's contribution only from the counter term above

loop:

 : $\quad -i\,\Pi^{(1)}(p^2) = \left(ig\mu^{\varepsilon/2}\right)^2 \frac{1}{2} \int \frac{d^dk}{(2\pi)^d} \frac{i}{(k^2-m^2)+i\varepsilon} \frac{i}{((k+p)^2-m^2+i\varepsilon)}$

Despite the arrows this is just a scalar propagator loop

The feynman rule for the vertex  is $-ig\mu^{\varepsilon/2}$

Let's now calculate $-i\,\Pi^{(1)}(p^2)$

$$-i\,\Pi^{(1)}(p^2) = -g^2\mu^\varepsilon \frac{1}{2} \int \frac{d^dk}{(2\pi)^d} \frac{i^2}{(k^2-m^2)+i\varepsilon} \frac{1}{((k+p)^2-m^2+i\varepsilon)}$$

|| Let's use Feynmann parametrization

$$= +\frac{g^2\mu^\varepsilon}{2} \int_0^1 dx \int \frac{d^dk}{(2\pi)^d} \frac{1}{\left(x[(k+p)^2-m^2+i\varepsilon] + (1-x)[k^2-m^2+i\varepsilon]\right)^2}$$

The denominator is D^2 such that

$$D = xk^2 + xp^2 - m^2x + i\varepsilon x + k^2 - m^2 + i\varepsilon - xk^2 + m^2x - i\varepsilon x + x2k\cdot p$$

$$\Rightarrow D = k^2 + 2x\cdot k\cdot p + x^2p^2 - x^2p^2 + xp^2 - m^2 + i\varepsilon$$

$$D = (k+xp)^2 - \underbrace{(x^2p^2 - xp^2 + m^2 + i\varepsilon)}_{\equiv \Delta}$$

Shift of variable: $\; l = k + xp \;\Rightarrow\; d^dk = d^dl$

$$\Rightarrow -i\,\Pi^{(1)}(p^2) = \frac{g^2\mu^\varepsilon}{2} \int_0^1 dx \int \frac{d^dl}{(2\pi)^d} \frac{1}{(l^2-\Delta)^2}$$

Let's now Wick rotate to Euclidean plane with $l^0 \to i l_E^0 \Rightarrow dl^0 \to i\, dl_E^0$
and $dl^i = dl_E^i \;\Rightarrow\; l^2 \to -l_E^2 \;\Rightarrow\; (l^2-\Delta)^2 \to (l_E^2+\Delta)^2$

$$\Rightarrow -i\,\Pi^{(1)}(p^2) = i\,\frac{g^2\mu^\varepsilon}{2} \int_0^1 dx \int \frac{d^dl_E}{(2\pi)^d} \frac{1}{(l_E^2+\Delta)^2}$$

Peskin eq. (7.85) gives us the integral:

$$\int \frac{d^dl_E}{(2\pi)^d} \frac{1}{(l_E^2+\Delta)^2} = (4\pi)^{-d/2} \frac{\Gamma(2-\frac{d}{2})}{\Gamma(2)} \Delta^{\frac{d}{2}-2}$$

|| $\Gamma(2) = 1$
$d = 6 - \varepsilon$

$$\Rightarrow \quad -i\Pi^{(1)}(p^2) = ig^2\mu^{\varepsilon}\frac{1}{2}\int_0^1 dx \; (4\pi)^{-3+\varepsilon/2}\;\Gamma(\tfrac{\varepsilon}{2}-1)\;\Delta^{1-\varepsilon/2}$$

$$-i\Pi^{(1)}(p^2) = i\frac{g^2}{2(4\pi)^3}\int_0^1 dx \;\left(\frac{4\pi}{m^2}\mu^2\right)^{\varepsilon/2}\Gamma\left(\frac{\varepsilon}{2}-1\right)\Delta\left(\frac{\Delta}{m^2}\right)^{-\varepsilon/2}$$

// We have added $1 = \left[(m^2)^{\varepsilon/2}\cdot(m^2)^{-\varepsilon/2}\right]^{-1}$

Let's now use the identity $\Gamma(\frac{\varepsilon}{2}-1) = (\frac{\varepsilon}{2}-1)^{-1}\,\Gamma(\frac{\varepsilon}{2})$

And $a^{\varepsilon/2} = e^{\frac{\varepsilon}{2}\ln a} \simeq 1+\frac{\varepsilon}{2}\ln(a) + \mathcal{O}(\varepsilon^2)$ for small ε

similarily $a^{-\varepsilon/2} = e^{-\frac{\varepsilon}{2}\ln(a)} \simeq 1-\frac{\varepsilon}{2}\ln(a) + \mathcal{O}(\varepsilon^2)$ for small ε

$$\Rightarrow \left(\frac{4\pi}{m^2}\mu^2\right)^{\varepsilon/2} \simeq 1+\frac{\varepsilon}{2}\ln\left[\frac{4\pi\mu^2}{m^2}\right] \quad \text{to the first order in } \varepsilon$$

$$\Rightarrow \left(\frac{\Delta}{m^2}\right)^{-\varepsilon/2} \simeq 1-\frac{\varepsilon}{2}\ln\left[\frac{\Delta}{m^2}\right] \quad \text{to the first order in } \varepsilon$$

Let's also use $(\frac{\varepsilon}{2}-1)^{-1} = -(1-\frac{\varepsilon}{2})^{-1} \simeq -(1+\frac{\varepsilon}{2})$

and the identity

$$\Gamma(\tfrac{\varepsilon}{2}) \simeq \frac{2}{\varepsilon} - \gamma_E + \mathcal{O}(\varepsilon)$$

$$\Rightarrow \quad -i\Pi^{(1)}(p^2) = -i\frac{g^2}{2(4\pi)^3}\int_0^1 dx \;\left(\frac{2}{\varepsilon}-\gamma_E\right)\left(1+\frac{\varepsilon}{2}\right)\left(1+\frac{\varepsilon}{2}\ln\left(\frac{4\pi\mu^2}{m^2}\right)\right)\left(1-\frac{\varepsilon}{2}\ln\left(\frac{\Delta}{m^2}\right)\right)\Delta$$

ε is small $\Rightarrow \left(1+\frac{\varepsilon}{2}\ln\left(\frac{4\pi\mu^2}{m^2}\right)\right)\left(1-\frac{\varepsilon}{2}\ln\left(\frac{\Delta}{m^2}\right)\right) \simeq 1+\frac{\varepsilon}{2}\ln\left(\frac{4\pi\mu^2}{m^2}\right)-\frac{\varepsilon}{2}\ln\left(\frac{\Delta}{m^2}\right)$.

and we will have $\varepsilon\to 0$, there is only one $\frac{2}{\varepsilon}$ term so we can take all ε^2 term to go to zero and only spare terms in ε before multiplying by $\left(\frac{2}{\varepsilon}-\gamma_E\right)$

$$\Rightarrow \quad -i\Pi^{(1)}(p^2) = -i\frac{g^2}{2(4\pi)^3}\int_0^1 \left(\frac{2}{\varepsilon}-\gamma_E\right)\left(1+\frac{\varepsilon}{2}\ln\left(\frac{4\pi\mu^2}{m^2}\right)-\frac{\varepsilon}{2}\ln\left(\frac{\Delta}{m^2}\right)\right)\Delta \; dx$$

Now we only spare terms that are of order $\frac{1}{\varepsilon}$ or $\varepsilon^0 = 1$ since $\varepsilon\to 0$

$$\Rightarrow \left(\frac{2}{\varepsilon}-\gamma_E\right)\left(1+\frac{\varepsilon}{2}\ln\left(\frac{4\pi\mu^2}{m^2}\right)-\frac{\varepsilon}{2}\ln\left(\frac{\Delta}{m^2}\right)\right) \simeq \frac{2}{\varepsilon}-\gamma_E+\ln\left[\frac{4\pi\mu^2}{m^2}\right]-\ln\left[\frac{\Delta}{m^2}\right]$$

$$\Rightarrow \quad -i\Pi^{(1)}(p^2) = -i\frac{g^2}{2(4\pi)^3}\left[\int_0^1 dx\left(\frac{2}{\varepsilon}-\gamma_E+\ln\left[\frac{4\pi\mu^2}{m^2}\right]\right)\Delta \; - \; m^2\int_0^1 dx\,\frac{\Delta}{m^2}\ln\left(\frac{\Delta}{m^2}\right)\right]$$

$\Delta = m^2 - x(1-x)p^2 + i\varepsilon \simeq m^2 - x(1-x)p^2$ sin $\varepsilon\to 0$ ($i\varepsilon$ - prescription was not necessary AFTER Wick rotation)

The integral $\int_0^1 dx\,(m^2 - x(1-x)p^2)$ gives $m^2 - (\frac{1}{2}-\frac{1}{3})p^2 = m^2 - \frac{1}{6}p^2$

$$\Rightarrow \quad -i\Pi^{(1)}(p^2) = -i\frac{g^2}{2(4\pi)^3}\left[\left(m^2-\frac{p^2}{6}\right)\left(\frac{2}{\varepsilon}-\gamma_E+\ln\left[\frac{4\pi\mu^2}{m^2}\right]\right) - m^2\int_0^1 dx\,\frac{\Delta}{m^2}\ln\left(\frac{\Delta}{m^2}\right)\right]$$

The divergent part is with $\frac{2}{\varepsilon} \Rightarrow -i\frac{g^2}{2(4\pi)^3}\left(m^2-\frac{p^2}{6}\right)\frac{2}{\varepsilon}$ is the divergent part

The counter terms are of form $i(\delta_Z p^2 - \delta m)$

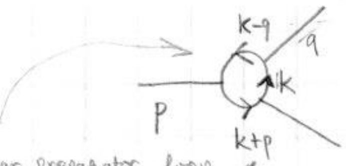
Which means that in the minimum subtraction scheme we get

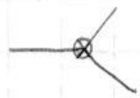
$$\delta_m^{MS} = -\frac{g^2}{2}\frac{m^2}{(4\pi)^3}\frac{2}{\varepsilon} \quad \text{and} \quad \delta_Z^{MS} = -\frac{g^2}{12(4\pi)^3}\frac{2}{\varepsilon} \quad \square$$

Exercise 7.9

Now we tackle the 3-point function.

The 3-point function in the first order is  , Now we introduce the 1-loop correction to the vertex (The Feynman rule is $-ig\mu^{\varepsilon/2}$ for the vertex)



with the counter term  $= -i\delta_g \mu^{\varepsilon/2}$

Scalar propagator loop (Arrow just to see momentum)

Feynmann rule straight from the lagrangian.

The loop integral is

$$\Gamma_3^{(1)}(p^2) = \Gamma^{(1)}(p^2) = \underbrace{(-ig\mu^{\varepsilon/2})^3}_{\text{From 3 vertex along the loop}} \int \frac{d^dk}{(2\pi)^d} \frac{i}{(k+p)^2 - m^2 + i\varepsilon} \frac{i}{k^2 - m^2 + i\varepsilon} \frac{i}{(k-q)^2 - m^2 + i\varepsilon}$$

Lets start by using Feynman parametrization $\frac{1}{ABC} = \int dx\,dy\,dz \frac{2!\,\delta(x+y+z-1)}{[xA+yB+zC]^3}$

$$\Rightarrow \Gamma^{(1)}(p^2) = 2(-ig\mu^{\varepsilon/2})^3 \int \frac{d^dk}{(2\pi)^d} \int dx\,dy\,dz\, \delta(x+y+z-1)\left[x(k+p)^2 + yk^2 + z(k-q)^2 + (x+y+z)(i\varepsilon - m^2)\right]^{-3}$$

Integrate over y to get

$$\Gamma^{(1)}(p^2) = 2(-ig\mu^{\varepsilon/2})^3 \int_0^1 dx \int_0^{1-x} dz \int \frac{d^dk}{(2\pi)^d}\left[x(k+p)^2 + (1-x-z)k^2 + z(k-q)^2 + i\varepsilon - m^2\right]^{-3}$$

The denominator D^3 can be simplified farther:

$$D = xk^2 + xp^2 + 2kpx + k^2 - xk^2 - zk^2 + zk^2 - z2kq + zq^2 + i\varepsilon - m^2$$

$$= k^2 + 2kpx - 2kqz + xp^2 + zq^2 + i\varepsilon - m^2 = k^2 + 2k(px - qz) + xp^2 + zq^2 + i\varepsilon - m^2$$

$$= k^2 + 2k(px - qz) + (px-qz)^2 - (px-qz)^2 + xp^2 + zq^2 + i\varepsilon - m^2 \quad \Big\| \quad p^2x^2 + q^2z^2 - 2xz\,pq = (px-qz)^2$$

$$= (k + px - qz)^2 + xp^2(1-x) + zq^2(1-z) + 2xz\,p\cdot q - m^2 + i\varepsilon$$

Let's do change of variable: $l = k + px - qz \Rightarrow dk = dl$

and $\Delta = -xp^2(1-x) - zq^2(1-z) - 2xz\,p\cdot q - m^2$

$\Delta > 0$ since $q^2 < 0$ and $p^2 < 0$

$$\Rightarrow D = l^2 - \Delta + i\varepsilon \quad \Rightarrow \quad \Gamma^{(1)}(p^2) = 2(-ig\mu^{\varepsilon/2})^3 \int_0^1 dx \int_0^{1-x} dz \int \frac{d^dl}{(2\pi)^d} \frac{1}{(l^2 - \Delta + i\varepsilon)^3}$$

Now wick rotate using $l^0 \to i l_E^0$ and $l^i \to l_E^i$

$\Rightarrow l^2 \to -l_E^2$ and $d^d l \to i d^d l_E$ and we drop the $i\varepsilon$ since denominator $\neq 0$ anywhere in Euclidean space.

$$\Rightarrow \Gamma^{(1)}(p^2) = 2i(-ig\mu^{\varepsilon/2})^3 \int_0^1 dx \int_0^{1-x} dz \int \frac{d^d l_E}{(2\pi)^d} \frac{-1}{(l_E^2+\Delta)^3}$$

From Peskin eq. (7.85) : $\displaystyle\int \frac{d^d l_E}{(2\pi)^d} \frac{1}{(l_E^2+\Delta)^n} = (4\pi)^{-d/2} \frac{\Gamma(n-d/2)}{\Gamma(n)} \Delta^{\frac{d}{2}-n}$

$$\Rightarrow \int \frac{d^d l_E}{(2\pi)^d} \frac{1}{(l_E^2+\Delta)^3} = (4\pi)^{-\frac{d}{2}} \frac{\Gamma(3-d/2)}{\Gamma(3)} \Delta^{\frac{d}{2}-3}$$

In our case $d = 6-\varepsilon$

$$\Rightarrow \int \frac{d^d l_E}{(2\pi)^d} \frac{1}{(l_E^2+\Delta)^3} \overset{d=6-\varepsilon}{=\!=} 4\pi^{-3+\varepsilon/2} \frac{\Gamma(\frac{\varepsilon}{2})}{\Gamma(3)} \Delta^{-\frac{\varepsilon}{2}} \qquad \Big\| \; \Gamma(3) = 2$$

$$\Rightarrow \quad = \frac{1}{2} \frac{1}{(4\pi)^3} (4\pi)^{\varepsilon/2} \Gamma(\varepsilon/2) \Delta^{-\varepsilon/2}$$

$$\Rightarrow \Gamma^{(1)}(p^2) = -2i(-ig\mu^{\varepsilon/2})^3 \frac{(4\pi)^{\varepsilon/2}}{2(4\pi)^3} \int_0^1 dx \int_0^{1-x} dz \, \Gamma(\varepsilon/2) \Delta^{-\varepsilon/2} \qquad \Big\| \; (-i)^3 = -i\cdot(-i)^2 = i$$

$$\Delta^{-\varepsilon/2} = \exp(-\tfrac{\varepsilon}{2} \ln\Delta) \simeq 1 - \tfrac{\varepsilon}{2}\ln\Delta + \mathcal{O}(\varepsilon^2)$$

$$\Gamma(\tfrac{\varepsilon}{2}) \simeq \tfrac{2}{\varepsilon} - \gamma_E + \mathcal{O}(\varepsilon)$$

$$(4\pi)^{\varepsilon/2} \mu^{3\varepsilon/2} = \mu^{\varepsilon/2} (4\pi\mu^2)^{\varepsilon/2} = \mu^{\varepsilon/2} \exp\left(\tfrac{\varepsilon}{2}\ln(4\pi\mu^2)\right) \simeq \mu^{\varepsilon/2}\left(1 + \tfrac{\varepsilon}{2}\ln(4\pi\mu^2) + \mathcal{O}(\varepsilon^2)\right)$$

We spare the $\mu^{\varepsilon/2}$ from approximation since the counter term is proportional to $\mu^{\varepsilon/2}$

$$\Rightarrow \Gamma^{(1)}(p^2) = \frac{g^3\mu^{\varepsilon/2}}{(4\pi)^3} \int_0^1 dx \int_0^{1-x} dz \left(1+\tfrac{\varepsilon}{2}\ln(4\pi\mu^2) + \mathcal{O}(\varepsilon^2)\right)\left(\tfrac{2}{\varepsilon} - \gamma_E + \mathcal{O}(\varepsilon)\right)\left(1 - \tfrac{\varepsilon}{2}\ln\Delta + \mathcal{O}(\varepsilon^2)\right)$$

After multiplying we drop terms of order $\mathcal{O}(\varepsilon)$ and higher since ε is small

$$\Rightarrow \Gamma^{(1)}(p^2) \simeq \frac{g^3\mu^{\varepsilon/2}}{(4\pi)^3} \int_0^1 dx \int_0^{1-x} dz \left[\tfrac{2}{\varepsilon} - \gamma_E + \ln(4\pi\mu^2) - \ln(\Delta)\right]$$

$$\int_0^1 dx \int_0^{1-x} dz = \int_0^1 dx\,(1-x) = (x - x^2 \tfrac{1}{2})\Big|_0^1 = 1 - \tfrac{1}{2} = \tfrac{1}{2}$$

$$\Gamma^{(1)}(p^2) = \frac{g^3\mu^{\varepsilon/2}}{2(4\pi)^3}\left(\underbrace{\tfrac{2}{\varepsilon}}_{\text{the divergent part}} - \gamma_E + \ln(4\pi\mu^2) - \int_0^1 dx \int_0^{1-x} dz \, \ln(\Delta)\right)$$

The three-point function is $-i\Gamma(p^2) = -i\Gamma^{(1)}(p^2) - i\delta_g\mu^{\varepsilon/2} - \underbrace{ig\mu^{\varepsilon/2}}_{\to\, ig \text{ as } \varepsilon\to 0}$ so doesn't have divergence

In the MS-scheme we recognise δ_g^{MS} as being term that exactly cancles the divergent part of $\Gamma^{(1)}(p^2)$

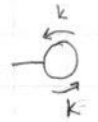
The divergent part of $\Gamma^{(1)}(p^2)$ is $\mu^{\varepsilon/2}\frac{g^3}{2(4\pi)^3}\frac{2}{\varepsilon} \equiv \Gamma^{(1)}_{\text{div}}(p^2)$

$$\Rightarrow \quad -i\,\Gamma^{(1)}_{\text{div}}(p^2) - i\,\delta_g^{MS}\mu^{\varepsilon/2} = 0$$

$$\Rightarrow \quad \delta_g^{MS}\mu^{\varepsilon/2} + \mu^{\varepsilon/2}\frac{g^3}{2(4\pi)^3}\frac{2}{\varepsilon} = 0$$

$$\Rightarrow \quad \delta_g^{MS} = -\frac{g^3}{2(4\pi)^3}\frac{2}{\varepsilon} \quad \blacksquare$$

Exercise 7.10

The 1-loop diagram for the 1-point function is  ← the momentum in loop is fixed by momentum conservation

The loop integral is $\mathcal{I} = \int \frac{d^dk}{(2\pi)^d}\frac{i}{(k^2-m^2)+i\varepsilon}$

Vertex gives $-ig\mu^{\varepsilon/2}$. the counter term gives $-i\delta_1$ (straigh from Lagrangion)

1-point function is given by $1+\Pi_1^{(1)}(p^2)-i\delta_1$ where $\Pi_1^{(1)}(p^2) = -ig\mu^{\varepsilon/2}\, I$

$$I = \int \frac{d^dk}{(2\pi)^d}\frac{i}{k^2-m^2+i\varepsilon}$$

let's do wick rotation: $k^0 \to ik_E^0$ and $k^i \to k_E^i$
$\Rightarrow d^dk \to i\,d^dk_E$ and $k^2 \to -k_E^2$

$$I = i^2\int \frac{d^dk_E}{(2\pi)^d}\frac{1}{(-k_E^2-m^2)}$$

// We can drop $i\varepsilon$ since there is no singularity in the Euclidean integral.

$$I = -i^2\frac{1}{(4\pi)^{d/2}}\frac{\Gamma(1-\frac{d}{2})}{\Gamma(1)}(m^2)^{\frac{d}{2}-1}$$

|| Here we used the integration formulae from Peskin (7.85)

$$I = +\frac{(4\pi)^{\varepsilon/2}}{(4\pi)^3}\frac{\Gamma(\frac{\varepsilon}{2}-2)}{1}(m^2)^{2-\varepsilon/2}$$

⟪ $d = 6-\varepsilon$, $\Gamma(1)=1$, $1-\frac{d}{2} = -2+\varepsilon/2$

$$\Gamma\left(\left(\tfrac{\varepsilon}{2}-1\right)-1\right) = \left(\tfrac{\varepsilon}{2}-2\right)^{-1}\Gamma\left(\tfrac{\varepsilon}{2}-1\right)$$

|| $\Gamma(\frac{\varepsilon}{2}-1) = (\frac{\varepsilon}{2}-1)^{-1}\Gamma(\frac{\varepsilon}{2})$

$$\Gamma\left(\tfrac{\varepsilon}{2}-2\right) = \left(\tfrac{\varepsilon}{2}-2\right)^{-1}\left(\varepsilon/2-1\right)^{-1}\Gamma\left(\tfrac{\varepsilon}{2}\right)$$

|| $(\varepsilon-2)^{-1} \approx -\frac{1}{2}$ and $(\varepsilon-1) \approx -1$ are valid approximation since $\varepsilon \ll 1$ and no need to use binomial approx since at the end we drop terms of $\mathcal{O}(\varepsilon)$ any ways

$$\Gamma\left(\tfrac{\varepsilon}{2}-2\right) \approx +\frac{1}{2}\Gamma\left(\tfrac{\varepsilon}{2}\right)$$

$$\Rightarrow \quad I = + \frac{(4\pi)^{\varepsilon/2}}{2(4\pi)^3} \Gamma\left(\frac{\varepsilon}{2}\right) m^4 \, m^{-\varepsilon}$$

$$\Rightarrow \quad -i g \mu^{\varepsilon/2} I = -i \frac{g (4\pi)^{\varepsilon/2} \mu^{\varepsilon/2}}{2(4\pi)^3} m^4 \, \Gamma\left(\frac{\varepsilon}{2}\right) m^{-\varepsilon} = \Pi_1^{(1)}$$

We now approximate: $(4\pi\mu)^{\varepsilon/2} \approx 1 + \frac{\varepsilon}{2} \ln(4\pi\mu)$ and $m^{-\varepsilon} \approx 1 - \varepsilon \ln(m)$
and $\Gamma\left(\frac{\varepsilon}{2}\right) \approx \frac{2}{\varepsilon} + \gamma_E$

$$\Rightarrow \Pi_{loop} = \Pi_1^{(1)} = -i \frac{g m^4}{2(4\pi)^3} \left(\frac{2}{\varepsilon} + \gamma_E\right)\left(1 + \frac{\varepsilon}{2} \ln(4\pi\mu)\right)\left(1 - \varepsilon \ln(m)\right)$$

$\varepsilon \to 0$ So terms of order $\mathcal{O}(\varepsilon)$ or higher vanish

$$\Pi_{loop} = -i \frac{g m^4}{2(4\pi)^3} \left(\frac{2}{\varepsilon} + \gamma_E + \ln(4\pi\mu) - 2\ln(m)\right)$$

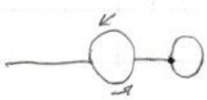
the divergent part has $\frac{2}{\varepsilon}$ and is the only term that needs to be countered by δ_1 in MS-scheme.

$$\Rightarrow \quad -\frac{i g m^4}{2(4\pi)^3} \frac{2}{\varepsilon} - i \delta_1^{MS} = 0$$

$$\Rightarrow \quad \underline{\delta_1^{MS} = -\frac{g m^4}{2(4\pi)^3} \frac{2}{\varepsilon}} \quad \square$$

This tadpole amplitude can be written as $\delta_1^{MS} = m^2 \delta_m^{MS}$, so there is connection with δ_m^{MS}.

If we allow the loops of 2-point function to also contain tadpole diagrams then there is no need for δ_1- counter term i.e. we allow Feynman diagram



to be allowed as 1-loop correction to the 2-point function. $\Rightarrow$ No need for distinct δ_1- counterterm.

For higher order corrections we allow tadpole contributions to similarily be in the 2-poin function corrections.